

large component sizes. The prototype underwent four iterations of high-level designs before one was finalised.

Scaling up to mass-production level was also not easy. He elaborates, “The whole process changes when you scale your product for commercial launch. You need to redesign the product keeping the scale in mind. This is called design for manufacturing (DFM).”

The units are being manufactured at a New Delhi-based factory. While the base helmet is being manufactured by one partner, electronic boards are being fabricated by another.

Finding the right manufacturer was also difficult. He explains, “While prototyping, unit sizes and order volumes are small. Manufacturers have to spend more time and effort on lower production. It is hard to convince them to cooperate for an innovation because changing the setup every time for an innovation is not what they make money out of. They make money on manufacturing a large number of units of the same design or products at a go.”

The innovation process of the features was difficult, too. “Complexity unravels as you keep delving into production,” points out Pathak.

A major strategy to reach out to the target audience is to work closely with the customers directly. Shellios participates in biker events where it showcases a Puros prototype and collects feedback.

Pathak says, “Our target segment includes citizens who own a 200cc or higher bike, who are willing to invest a reasonable price on a helmet for safety as well as comfort.” The price tag will be announced soon. Shellios will first roll out the helmets and later deploy the mobile application that is currently under development.

For quality assurance, the team is currently getting the product certified by Bureau of Indian Standards (BIS).

Regarding marketing strategies, Pathak shares, “We will not be very media-heavy. However, we will be active with our customer interactions. We will participate in biker fests during winter months and get direct feedbacks. Digital marketing will also be an important operation.”

Shellios helmets will be distributed through online platforms like Flipkart, Amazon and automotive-specific sites. These will also be sold through offline channels via selected auto accessory stores.

Team Shellios has filed for a patent for Puros. It plans to strengthen its market presence steadily, taking one step at a time.

Bandicoot: Made In India Robot Replacing Manual Scavenging

One of the biggest problems in India is the profession of manual scavenging. Although laws have been made to stop the practice, human scavengers are forced to enter and clean the manholes with their hands even today—a major reason being a lack of proper technological solutions to take their place.

Lack of proper education and job opportunities drive most unprivileged citizens towards such jobs. Harmful gases, toxic fluids and risky environment can prove life-altering or even fatal for them. To address these

issues, Genrobotic Innovations, a Thiruvananthapuram-based robotics firm created an intelligent, manhole-cleaning robot called Bandicoot.

Owing to a common interest in robotics and automation, college mates Vimal Govind M.K. (chief executive officer), Rashid K. (chief marketing officer), Arun George (chief administrative officer) and Nikhil N.P. (chief technology officer) co-founded GenRobotics in 2015. An accident in Kozhikode district where three men lost their lives after getting trapped inside a manhole triggered their focus on creating a solution that would ensure no humans needed to enter manholes. In 2016, they got the opportunity to meet government officials of the state of Kerala, and discussed various sanitation issues and possible solutions.

With the help of Kerala Startup Mission and further support from Kerala Water Authority (KWA), the trio started research on the subject. Kerala Startup Mission aided them with space and infrastructure to develop the project. KWA provided them with data on various manholes across cities in Kerala, including manhole dimensions, inner environment, potential risks and so on. After in-depth research and analysis, they started developing the robot.

In 2017, GenRobotics was officially registered as a firm, and the first prototype of Bandicoot robot was tested in an in-lab environment. After rigorous tests and improvements, GenRobotics finally deployed their first commercial model of Bandicoot in February 2018.



GenRobotics team with Bandicoot

Working technology

Explaining the functionality of Bandicoot, George says, “To get the job done, the robot goes inside the manhole and mimics all the actions of a human scavenger. Bandicoot has two structures. One is the